IGCSE 0607 Practice Questions — E1.16

Exponential growth and decay

Total Questions: 21 — **Total Marks:** 114 Marks

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1 A population of bacteria increases exponentially at a constant rate of $p\%$ per hour.
At 09 : 00, the population is 8 000.
At 12 : 00 on the same day, the population is 27 000.

(a) **(a)** Work out the value of p.

$p =$ ..... **[3]**

(b) Work out the population of the bacteria at 14 : 00 on the same day.

..... **[2]**

2 **(a)** The value of a painting increases exponentially by $r\%$ each year.
After 2 years, the total percentage increase in the value of the painting is 44%.
Work out the value of r.

$r =$ ..... **[3]**

(b) The value of a rare stamp increases exponentially.
Its value doubles every 15 years.
The current value of the stamp is \$80.

Work out the value of the stamp after 45 years.

% ..... **[2]**

3 A new car is bought for \$24 000. The value of the car depreciates exponentially at a rate of 8% per year.

Calculate the value of the car after 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**

4 The population of a city is 125 000.
The population increases exponentially at a rate of 1.8% per year.
Calculate the population of the city after 8 years.
Give your answer correct to the nearest integer.

..... **[3]**

5 A new car is bought for \$26000. The value of the car depreciates exponentially at a rate of 15% each year.

Work out the value of the car after 4 years.
Give your answer correct to the nearest dollar.

% ..... **[3]**

6 A rare stamp is bought for \$450. Its value increases exponentially at a rate of 6.5% per year. Calculate the value of the stamp after 12 years.

\$ ..... **[3]**

7 A machine is bought for \$32 000. It depreciates exponentially at a rate of 12% each year.

Calculate the value of the machine after 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**

8 The population of a town is 65000.
The population increases exponentially at a rate of 2.8% per year.

(a) Calculate the expected population of the town after 5 years.
Give your answer correct to the nearest integer.

..... **[2]**

(b) Calculate the overall percentage increase in the population over the 5 years.
Give your answer correct to 1 decimal place.

..... % **[2]**

9 The population of a town at the start of 2020 was 42 000.
The population increases at a constant rate of 2.5% each year.

(a) Calculate the population of the town at the start of 2026.
Give your answer correct to the nearest hundred.

..... **[2]**

(b) Calculate the percentage increase in the population from the start of 2020 to the start of 2026.
Give your answer correct to 1 decimal place.

..... % **[2]**

10 A population of bacteria in a dish grows exponentially.

(a) Initially, there are 1500 bacteria in the dish.
The population increases by 8% every hour.
Work out the population of bacteria after 12 hours.
Give your answer correct to the nearest whole number.

..... **[2]**

(b) Work out the least number of complete hours it will take for the initial population of 1500 bacteria to be greater than 3000.

..... hours **[3]**

11 The population of a town was 85 000 on 1 January 2010.
The population increases exponentially at a constant rate of $r\%$ per year.
On 1 January 2022, the population was 112 000.

(a) **(a)** Calculate the value of r, correct to 3 significant figures.

$r =$ ..... **[3]**

(b) Assuming the rate of increase remains the same, calculate the expected population on 1 January 2030.
Give your answer correct to the nearest thousand.

..... **[2]**

12 A car is purchased for \$28 000. Its value depreciates exponentially at a rate of $r\%$ each year. After 4 years, its value is \$16 150.

(a) Calculate the value of r.
Give your answer correct to 3 significant figures.

$r =$ ..... **[3]**

(b) The value of the car continues to depreciate at the same rate.
Calculate the value of the car after a further 3 years.
Give your answer correct to the nearest dollar.

\$ ..... **[2]**

13 A machine costs \$45000. Its value depreciates exponentially at a rate of $x\%$ per year. After 4 years, the value of the machine is \$26000.

(a) Calculate the value of x.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

(b) The machine will be replaced when its value first falls below \$10000.
Find the number of complete years before the machine is replaced.

..... years **[3]**

14 **(a)** The population of a city is 125 000.
It grows exponentially at a rate of 2.4% per year.

Calculate the expected population after 10 years.
Give your answer correct to the nearest hundred.

..... **[3]**

(b) The population of another city was 85 000 in the year 2015.
By 2023, the population had decreased exponentially to 68 000.

Calculate the annual percentage decrease.

..... % **[3]**

15 A delivery van is bought new for \$36 000. Its value depreciates exponentially by 14% each year.

(a) Calculate the value of the van after 4 years.
Give your answer correct to the nearest dollar.

\$ ..... **[2]**

(b) Calculate the number of complete years it takes for the value of the van to first fall below \$10 000.

..... years **[4]**

16 Consider the depreciation of two different vehicles over time.

(a) A car is purchased for \$28 500. Its value depreciates exponentially at a rate of 12% per year. Calculate the value of the car after 5 years, giving your answer correct to the nearest dollar.

\$ ..... **[3]**

(b) A van, initially purchased for \$35 000, depreciates exponentially to a value of \$15 200 after 4 years.
Calculate the annual percentage rate of depreciation for this van.

..... % **[4]**

17 The mass, M grams, of a radioactive substance decays exponentially at a rate of $r\%$ per day.
After 2 days, the mass is 80 grams.
After 5 days, the mass is 40.96 grams.

(a) Calculate the value of r.

$r =$ ..... **[3]**

(b) Find the initial mass of the substance.

..... grams **[2]**

(c) Calculate the number of complete days from the start when the mass is first less than 10 grams.

..... days **[3]**

18 A radioactive substance decays exponentially.

(a) The mass of the substance decreases by 12% every day.
The initial mass is 500 grams.

Calculate the mass remaining after 1 week (7 days).

..... g **[2]**

(b) After d complete days, the mass of the substance falls below 50 grams.

Find the smallest integer value of d.

$d =$ ..... **[3]**

(c) A different radioactive substance decays exponentially such that its mass halves every 15 days.

Calculate the percentage decrease in its mass each day.

..... % **[3]**

19 The mass, M grams, of a radioactive substance remaining after t hours is modelled by the formula

$$M = M_0 \times k^{-t}$$

where M_0 and k are positive constants.
At $t = 2$, the mass is 256 grams.
At $t = 5$, the mass is 108 grams.

(a) Show that $k = \frac{4}{3}$.

[3]

(b) Find the value of M_0.

$M_0 =$ ..... **[2]**

(c) Calculate the value of t when the mass of the substance has decreased to 10% of its initial mass M_0.
Give your answer correct to 2 decimal places.

$t =$ ..... **[3]**

20 Alex and Ben each invest \$4000 for 10 years. Alex invests his \$4000 in an account paying compound interest at a rate of 4.5% per year.
Ben invests his \$4000 in an account paying compound interest at a rate of r% per year.

(a) Calculate the total amount in Alex's account after 10 years.

\$ ..... **[2]**

(b) After 10 years, Ben has \$500 more than Alex.
Calculate the value of r.

$r =$ ..... **[4]**

(c) A different investment of \$2500 grows exponentially. It doubles in value every 8 years.
Calculate its value after 20 years.

\$ ..... **[2]**

21 The population of a species of insect increases exponentially.
Initially, there are 2500 insects.
After 6 weeks, there are 9100 insects.

(a) Calculate the percentage increase in the population per week.

..... % **[4]**

(b) Calculate the insect population after 10 weeks, giving your answer correct to the nearest 100.

..... **[2]**

(c) Find the number of complete weeks it takes for the initial population to increase to over 50 000.

..... weeks **[3]**